İstanbul GEDİK University

CERTIFICATE

“YUSUF İSLAM BUDAK”

This document is awarded to the individual who has successfully completed the

“Data Science and Machine Learning Training with Python”

The training program lasted 60 hours, and participant was awarded certificates.

BARCODE NUMBER: UN_041057[redacted]11764

TURKISH ID NUMBER: [redacted]

ISSUED DATE: 20.09.2024

PROF. DR. AHMET KESİK

RECTOR

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TRANSCRIPT and CONTENT DOCUMENT

TURKISH ID NUMBER:	[redacted]	**ISSUED DATE:**	20.09.2024
NAME SURNAME:	YUSUF İSLAM BUDAK	**TOTAL TIME:**	60
EDUCATION NAME:	Data Science and Machine Learning Training with Python		

Introduction of Python Program	Usage areas
Downloading and Installing	Basic Operators
Assignment Operators	Arithmetic Operators
Logical Operators	Data Types
Accessing Variable Elements by Variable Type	Conversions Between Variable Types
Receiving Transaction Outputs	Basic Text Functions
Predefined Functions	Function Assignment and Writing Functions
Local and Global Variables	Use of If Statement
If and Else Relationship	Relationship between If, Else and Elif
loops	For Loop
While Loop	Data processing
Module Concept and Module Operation	Installing the Numpy Module
Random Data Generation	Generating Data from Normal Distribution

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Generating Integer Data	Creating Numpy Strings
Properties of Numpy Strings	Reshaping Operations
Concatenating Strings	Splitting Procedures for Strings
Sorting Function on Strings	Access to Elements via Indexes
Range Selection Procedures	Matrices and Two-Dimensional Range Selection Procedures
Performing Subset Operations	Mathematical Operations on Strings
Mathematical Operations with Functions	Basic Mathematical and Statistical Operations
System of Equations with Two Unknowns and Numpy Module	Installing the Pandas Module
Changing Index Values	Creating Lists with the Help of Dictionaries
Creating a New Series by Combining Two Series	Element Operations in Pandas Module
Data Frame	Data Frame Element Operations
Selecting Variables with Loc and Iloc Functions	Conditional Element Selection Functions
Combining Operations in Data Frames	Matching Procedures in Data Frames
Importing Data Importing and Transferring Data Externally	Grouping Operations
Aggregation Function	Filtering Procedures
Repeat Commands	Sorting Operations
Conversion Functions	Pivot Tables
Data Visualization	Categorizing Data
Barplot Charts	Installing the Seaborn Module
Creating Scatter Plots with the Seaborn Module	Creating Scatterplots with the Seaborn Module

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Histograms and Similarities in Pandas and Seaborn Module	Obtaining Scatter Diagrams According to Different Breakdowns
Basic Statistical Information	A Look at Survey Data
Population and Sample	Descriptive and Inferential Statistics
Variable, Fixed Parameter	Dependent Independent Variable
Measurement Levels	Classification (Nominal) Scale
Ordinal Scale	Interval Scale
Basic Statistical Concepts	Average
mode	Median
Standard deviation	Variance
correlation	covariance
Normal distribution	Hypothesis Tests
Stages of Hypothesis Testing	Introducing the Interface of the Program
Menus in the Program	Saving and Opening the Work File
Variable View and Data View in Program	Creating a Variable and Variable-Specific Settings
Entering Data Effects	Assigning Category Names to Categorical Variables
Assigning the Measurement Level of the Variable	Data Entry for Assigned Variable
Converting Continuous Variables to Categorical Variables	Visual Grouping
Observation and Variable Access	Observing and Adding Frequencies
Sorting Variables Ascending and Descending	Selecting Observations as a Subset with Specific Conditions
Weighting Observations	Combining Categorical Variables by Recoding

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Creating New Variables from Mathematical Operations	Scale Calculations
Analysis of Reliability of Survey Data	Creating Crosstabs
Creating Frequency Tables	Obtaining Descriptive Statistics
Establishing Hypotheses Regarding Normal Distribution Tests	Normal Distribution Test
Test of Group Means	One Sample t Test
Independent Sample t Test	Nonparametric Independent Sample Test (Mann Whitney U)

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